Design of a Modern Textile Mill – Evaluation of Potential Businesses

March 2023

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Content

Quality of Ugandan Cotton

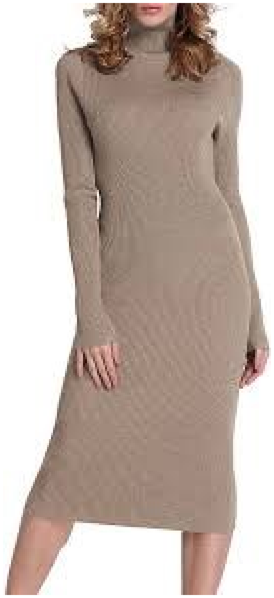
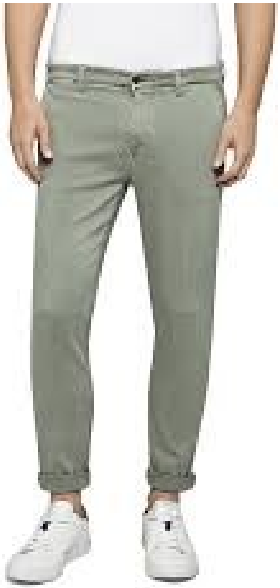
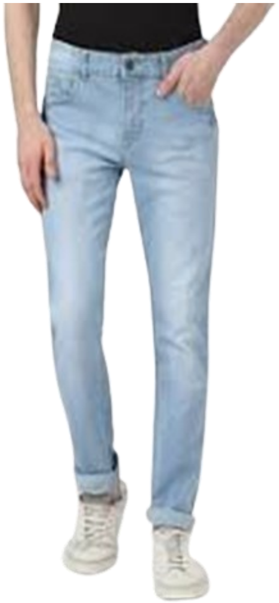
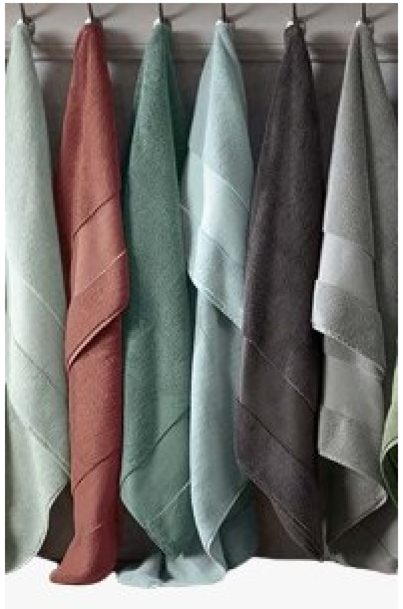
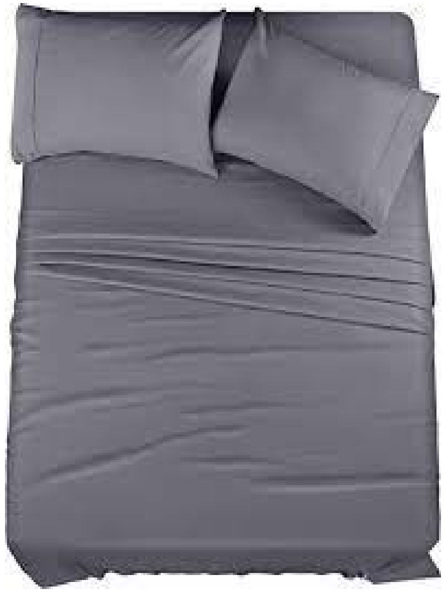
Fiber Properties of Ugandan Cotton

Type of Cotton	Length (mm)	Strength (g/tex)	Micronaire
UCOP	28-29	28-32	4.0 - 4.2
UCOB	28-29	30-33	3.7 – 4.3

Data Source: International Cotton Advisory Committee

- As per the technical parameters, Ugandan cotton is best suited to produce carded and combed yarn upto Ne 34s. For the production of finer yarns, fiber length of more than 29 mm is desirable.
- Yarns in the range of Ne 36s to Ne 40s can also be manufactured provided consistent 29 mm fiber length is available throughout the year. To avoid any manufacturing and quality issues in the long run, it is suggested that the core yarn product portfolio should be up to Ne 34s.

Potential Products (From Cotton Yarn up to Ne 34s)

Garment			Home Textile	
Knitted Dresses, Tops and T-shirts	Woven Bottom Wear (Trousers, Shorts and Skirts)	Denim	Towel	Bed Sheets (up to 180 TC*, low end)
				

*Thread Count which is sum of End Per Inch (EPI) and Picks Per Inch (PPI)

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Project Ballpark Numbers for Potential Products (as per Minimum Economic Size)

Garment

Knitted Dresses, Tops and T-Shirts – Project Overview

Process Flow: Spinning → Yarn Dyeing → Knitting → Fabric Processing → Garmenting

Particular	Spinning (Ne 24s and 30s)	Yarn Dyeing	Knitting	Fabric Processing	Garmenting	Total
Capacity	15,150 Spindles	2 tpd	48 Knitting Machines and Soft Flow Dyeing Machines		1,424 M/c (1 shift) 712 M/c (2 shifts)	
Production	10 tpd	1.6 tpd	9.6 tpd		21,300 pieces/day*	
External Sales	NIL	NIL	15%		100%	
Built-up Area (sqm)	8,200	200	15,500		21,350	**45,250**
Manpower Required	182	14	192		3,131	**3,519**
Investment (US$ mn.)	**13**	**0.5**	**17**		**12 (1 shift) 10 (2 shifts)**	**42 (1 shift) 40 (2 shifts)**
Peak Revenue Potential – 4h Year (US$ mn.)			**3**		**41**	**44**
Power (MW)						**4.0**
Water (MLD)						**1.4**
Cotton Requirement – 4th Year (tons/annum)						**3,977**
Working Capital Requirement – 4th Year						**18**

All the numbers are estimated based on ballpark figures

*Production - Pcs/day/machine at maximum efficiency (55%): 15

Factory Working Considered:

Textiles: 350 days, 8 hours x 3 shifts

Garment: 300 days, 8 hours x 1 shift / 2 shifts

Knitted Dresses, Tops and T-Shirts – Business Ramp-up

			Y1	Y2	Y3	Y4
Ramp-up	Spinning	Utilization	40%	51%	68%	95%
		Efficiency	75%	80%	85%	95%
	Knitting	Utilization	34%	43%	57%	80%
	Garment	Utilization (out of 300 days)	50%	60%	80%	100%
		Efficiency	35%	40%	45%	55%
Production	Yarn	tpd	4.3	5.4	7.1	10.0
	Fabric	tpd	4.1	5.2	6.8	9.6
	Garments	Pieces/day	7,159	9,818	14,114	21,333
Cotton Consumption		Tons/annum	1,690	2,138	2,834	3,977
Employment		Number	1,868	2,222	2,822	3,519
Revenue		US$ mn.	16	22	30	44
Revenue Growth				32%	39%	46%
Working capital		US$ mn.	6.4	8.8	12	17.6

Woven Bottom Wear (Trousers, Shorts and Skirts) – Project Overview

Process Flow: Spinning → Weaving → Fabric Processing → Garmenting

Particular	Spinning (Ne 20s)	Weaving	Fabric Processing	Garmenting	Total
Capacity	16,670 Spindles	110 Looms	70,000 meter/day	4,524 M/c (1 shift) 2,262 M/c (2 shift)	
Production	15 tpd	63,000 meter/day	58,000 meter/day	45,230 pieces/day*	
External Sales	NIL	NIL	15%	100%	
Built-up Area (sqm)	8,210	8,050	32,550	67,860	**1,16,670**
Manpower Required	200	138	294	9,953	**10,585**
Investment (US$ mn.)	**14**	**8**	**10**	**39 (1 shift) 32 (2 shifts)**	**72 (1 shift) 65 (2 shifts)**
Peak Revenue Potential – 4th Year (US$ mn.)			**11**	**95**	**106**
Power (MW)					**9.0**
Water (MLD)					**1.4**
Cotton Requirement – 4th Year (tons/annum)					**6,090**
Working Capital Requirement – 4th Year					**42**

All the numbers are estimated based on ballpark figures

*Production - Pcs/day/machine at maximum efficiency (55%): 10

Factory Working Considered:

Textiles: 350 days, 8 hours x 3 shifts

Garment: 300 days, 8 hours x 1 shift / 2 shifts

Woven Bottom Wear (Trousers, Shorts and Skirts) – Business Ramp-up

			Y1	Y2	Y3	Y4
Ramp-up	Spinning	Utilization	39%	49%	65%	96%
		Efficiency	75%	80%	85%	95%
	Processing	Utilization	34%	43%	57%	84%
	Garment	Utilization (out of 300 days)	50%	60%	80%	100%
		Efficiency	35%	40%	45%	55%
Production	Yarn	tpd	6	8	10	15
	Fabric	Meter/day	23,800	30,100	39,900	58,800
	Garments	Pieces/day	14,636	20,000	29,864	45,231
Cotton Consumption		Tons/annum	2,465	3,118	4,133	6,090
Employment		Number	5,366	6,402	8,469	10,585
Revenue		US$ mn.	41	52	72	106
Revenue Growth				29%	37%	48%
Working capital		US$ mn.	16	21	29	42

Denims – Project Overview

Process Flow: Spinning → Weaving → Fabric Processing → Garmenting

Particular	Spinning		Weaving	Fabric Processing	Garmenting	Total
	Ring Spinning (Ne 20s)	Open End (Ne 16s)				
Capacity	15,350 Spindles	480 Rotors	50 Looms	35,000 meter/day	1,130 M/c (1 shift) 565 M/c (2 shifts)	
Production	14 tpd	3 tpd	31,500 m/day	29,400 meter/day	22,615 pieces/day*	
External Sales	NIL	NIL	NIL	15%	100%	
Built-up Area (sqm)	7,680	660	3,500	14,700	16,965	**43,505**
Manpower Required	182	10	63	147	2,489	**2,891**
Investment (US$ mn.)	**13**	**2**	**4**	**5**	**14 (1 shift) 12 (2 shifts)**	**38 (1 shift) 36 (2 shifts)**
Peak Revenue Potential – 4th Year (US$ mn.)				**6**	**54**	**60**
Power (MW)						**4.0**
Water (MLD)						**1.5**
Cotton Requirement – 4th Year (tons/annum)						**6,688**
Working Capital Requirement – 4th Year						**24**

All the numbers are estimated based on ballpark figures

*Production - Pcs/day/machine at maximum efficiency (55%): 20

Factory Working Considered:

Textiles: 350 days, 8 hours x 3 shifts

Garment: 300 days, 8 hours x 1 shift / 2 shifts

Denims - Business Ramp-up

			Y1	Y2	Y3	Y4
Ramp-up	Spinning	Utilization	39%	50%	66%	97%
		Efficiency	75%	80%	85%	95%
	Processing	Utilization	34%	43%	57%	84%
	Garment	Utilization (out of 300 days)	50%	60%	80%	100%
		Efficiency	35%	40%	45%	55%
Production	Yarn	tpd	7	9	11	17
	Fabric	Meter/day	11,900	15,050	19,950	29,400
	Garments	Pieces/day	8,273	10,909	15,545	22,615
Cotton Consumption		Tons/annum	2,707	3,424	4,538	6,688
Employment		Number	1,640	1,882	2,372	2,891
Revenue		US$ mn.	24	31	41	61
Revenue Growth				28%	35%	47%
Working capital		US$ mn.	10	12	17	24

Home Textiles

Towels – Project Overview

Process Flow: Spinning → Yarn Dyeing → Weaving → Fabric Processing → Cutting & Stitching

Particular	Spinning		Yarn Dyeing	Weaving	Fabric Processing	Cutting & Stitching	Total
	Ring Spinning (Ne 20s)	Open End (Ne 16s)					
Capacity	7,660 Spindles	480 Rotor	2 tpd	28 Looms	12 tpd	Automatic and Semi-automatic Machines	
Production	8 tpd	3 tpd	1.6 tpd	10 tpd		25,935 pieces/day	
Built-up Area (sqm)	3,800	700	200	5,600			**10,300**
Manpower Required	90	10	14	196		680	**990**
Investment (US$ mn.)	**7**	**2**	**0.5**	**7**		**3**	**20**
Peak Revenue Potential - 4th Year (US$ mn.)							**22**
Power (MW)							**2.0**
Water (MLD)							**1.0**
Cotton Requirement – 4th Year (tons/annum)							**4,133**
Working Capital Requirement – 4th Year							**9**

All the numbers are estimated based on ballpark figures

Factory Working Considered: 350 days, 8 hours x 3 shifts

Towels – Business Ramp-up

			Y1	Y2	Y3	Y4
Ramp-up	Spinning	Utilization – Ring Spinning	46%	58%	77%	97%
		Utilization – Open End	24%	31%	41%	97%
		Efficiency	75%	80%	85%	95%
	Processing	Utilization	34%	43%	57%	82%
	Garment	Utilization (out of 300 days)	50%	60%	80%	100%
		Efficiency	35%	40%	45%	55%
Production	Yarn	tpd	4.3	5.4	7.2	10.4
	Fabric	tpd	4.1	5.2	7.0	10.0
	Garments	Pieces/day	10,309	12,960	18,556	25,936
Cotton Consumption		Tons/annum	1,716	2,170	2,876	4,133
Employment		Number	550	606	765	990
Revenue		US$ mn.	9	12	17	23
Revenue Growth				26%	43%	40%
Working capital		US$ mn.	4	5	7	9

Bed Sheets – Project Overview

Process Flow: Spinning → Weaving → Fabric Processing → Cutting & Stitching

Particular	Spinning (Ne 30s)	Weaving	Fabric Processing	Cutting & Stitching	Total
Capacity	21,675 Spindles	128 Looms	70,000 meter/day	Automatic and Semi-automatic Machines	
Production	16.7 tpd	57,400 meter/day		13,670 sets/day	
Built-up Area (sqm)	10,900	24,600			**45,500**
Manpower Required	261	420		123	**804**
Investment (US$ mn.)	**19**	**37**			**56**
Peak Revenue Potential -4th Year (US$ mn.)					**29**
Power (MW)					**7.0**
Water (MLD)					**1.5**
Raw Material Requirement(Cotton)					**8,125**
Working Capital Requirement					**12**

All the numbers are based on ballpark figures

Factory Working Considered: 350 days, 8 hours x 3 shifts

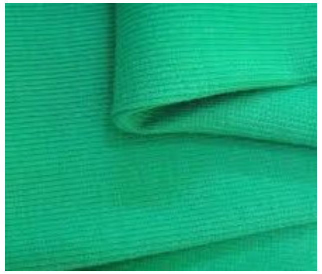
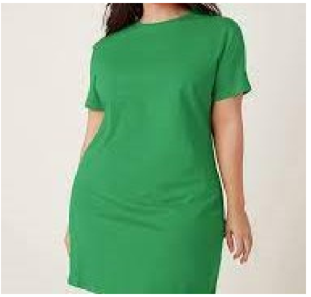
Bed Sheets – Business Ramp-up

			Y1	Y2	Y3	Y4
Ramp-up	Spinning	Utilization	40%	51%	68%	97%
		Efficiency	75%	80%	85%	95%
	Processing	Utilization	34%	43%	57%	82%
	Garment	Utilization (out of 300 days)	50%	60%	80%	100%
		Efficiency	35%	40%	45%	55%
Production	Yarn	tpd	6.9	8.8	11.6	16.7
	Fabric	Meter/day	23,800	30,100	39,900	57,400
	Garments	Pieces/day	11,598	13,255	14,911	13,667
Cotton Consumption		Tons/annum	3,374	4,268	5,657	8,125
Employment		Number	443	480	559	804
Revenue		US$ mn.	21	24	27	29
Revenue Growth				14%	13%	7%
Working capital		US$ mn.	8	10	11	12

Note to Revenue Projections

- The key requirements to meet the revenue projections will be:
    - Recruitment and training of manpower as per the capacity ramp-up
    - Availability of technical staff and middle management including expats
    - Deployment of modern technology machinery with automation
    - Establishment of strong raw material and buyer linkages
    - Setting-up of key support departments of product development and testing
    - Sufficient availability of working capital
    - Availability of required uninterrupted power and water

Value Addition Advantage of Forward Integration

	Raw Cotton		24s Ring Spun Yarn		180 GSM Single Jersey Knitted Fabric (dyed)		Women's Dress
							
	US$ 2.24 / kg		**US$ 3.29 / kg**		**US$ 5.89 / kg**		**US$ 6.5 / piece**
Equivalent Production & Value	1 ton US$ 2,240	=	0.88 ton US$ 2,895	=	0.79 ton US$ 4,653	=	1,756 piece US$ 11,414
Value Addition →			**29%**		**61%**		**145%**

Total value addition opportunity from fibre to garment is more than 400%
(US$ 11,000 worth of garment exports from US$ 2,200 worth of cotton)

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Technology Comparison

	Spinning	Yarn Dyeing	Fabric Production		Garmenting
			Knitting/Weaving	Processing	
Knitted Dresses, Tops and T-shirts	Ring Spinning (Ne 24s and 30s)	HTHP Dyeing	Circular Knitting	Soft Flow Dyeing	Stitching Machines
Woven Bottom Wear	Ring Spinning (Ne 20s)	Not Required	Airjet Looms	Continuous Fabric Processing Range	Stitching Machines
Denim	Ring Spinning (Ne 20s) and Open End (Ne 16s)	Not Required	Airjet Looms	Slasher Dyeing Fabric Processing Range	Stitching Machines (Heavy Duty and Speciality)
Towel	Ring Spinning (Ne 20s) and Open End (Ne 16s)	HTHP Dyeing	Airjet Pile Looms	Soft Flow Dyeing	Automatic Cutting and Side Hemming Line
Bedsheet	Ring Spinning (Ne 30s and finer count)	Not Required	Wider Width Airjet Looms	Continuous Fabric Processing Range	Automatic Cutting and Side Hemming Line

Summary

Category	Investment (US$ Mn.)		Peak Revenue Potential -4th Year (US$ Mn.)	Peak Revenue to Investment	Working Capital (US$ Mn.)	Production	Peak Cotton Required (Tons/ annum)	Employment	Built-up Area (sqm)	Connected Load (MW)	Water Requirement (MLD)
	Scenario 1	Scenario 2									
Knitted Dresses, Tops & T-shirts	42	40	44	1.1x	18	21,300 pieces/day	3,977	3,519	45,250	4.0	1.4
Woven Bottom Wear	72	65	106	1.5x	42	45,230 pieces/day	6,090	10,585	1,16,670	9.0	1.4
Denims	38	36	60	1.6x	24	22,615 pieces/day	6,688	2,891	43,505	4.0	1.5
Towels	20	-	22	1.1x	9	25,935 pieces/day	4,133	990	10,300	2.0	1.0
Bedsheets	56	-	29	0.5x	12	13,670 sets/day	8,125	804	45,500	7.0	1.5

Particular	Scenario 1	Scenario 2
Textiles	350 days, 8 hour x 3 shifts	350 days, 8 hour x 3 shifts
Garmenting	300 days, 8 hour x 1 shifts	300 days, 8 hour x 2 shifts

Summary

- Based the minimum viable capacity and optimum utilization, the smallest project would be of towel manufacturing (US$ 20 Mn.) and the biggest project will be for woven bottom wear (US$ 80 Mn.).
- The key capacity deciding section is fabric processing in each of the projects. There is a possibility to reduce investment in yarn, fabric (knitting or weaving) or cut & sew to reduce the investment but that will lead to under utilization of processing and increase the cost of overheads.
- Bedsheet project has minimum revenue to investment ratio (0.5x) because the product that can be manufactured is low end (180 TC bedsheets). The project profitability will hence be lowest.
- In each of the project, cut & sew department employs maximum workforce (75% - 90%). In garment projects (knits, wovens and denims), there is an option to reduce the cut & sew activity in line with labour availability. As of now, it is considered that 85% of the fabric will be converted into garments and balance 15% will be sold externally.
- It is not advisable to set-up a manufacturing facility for two different products because of the following reasons:
    - Denim and towel projects are standalone projects which do not have any process or product alignment with other three products.
    - Bedsheets and woven bottom wear are based on woven fabrics but the loom types and cut & sew technology for both are very different. In addition, the marketing base and buyers are totally different for both the projects.
    - Knitted garment and woven garments can have some synergy in cut & sew operations, however the fabric production and processing is entirely different.

Comparative Assessment of Projects

Challenges / Products	Technical Challenge	Volume Level	Skill Requirement	Imported Content (Spandex)	Product Development Challenge	Product Variety Limitation	Fabric Sale Constraint
Knitted Dresses, Tops and T-shirts	Positive	Positive	Moderate	Moderate	Moderate	Positive	Positive
Woven Bottom Wear	Positive	Negative	Moderate	Moderate	Positive	Positive	Positive
Denim	Negative	Moderate	Negative	Negative	Negative	Moderate	Positive
Towel	Positive	Positive	Moderate	Positive	Positive	Negative	Negative
Bedsheet	Moderate	Negative	Moderate	Positive	Moderate	Negative	Negative

Positive Moderate Negative Recommended Product

Recommendations

Product	Rationale
Priority 1 Knitted Dresses, Tops and T-shirts (10 tpd)	• Target product and fabric have high regional and global demand • Target product is not a technically complex product • The product range could cover basic commodity items such as t-shirts to value added high end products such as premium dresses
Priority 2 Towels (10 tpd)	• No regional competition • Target product is not a technically complex product • No import dependency for materials • High level of automation is possible in cut & sew department
Priority 3 Woven Bottom Wear (Trousers, Shorts and Skirts) (58,000 meter/day)	• Target product and fabric have high regional and global demand • Target product is not a technically complex product • The product range could cover variety of categories (trousers, shorts, skirts, chinos, bermuda shorts, cargo pants, etc.)

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